

Dynamic Optimization

Numerical Algorithms: From Static Solvers to RL

Numerical Algorithms Course

February 20, 2026

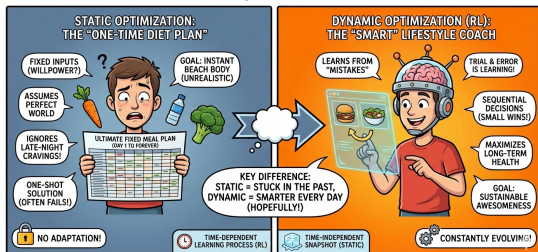
Static vs. Dynamic Optimization

Key Perspectives:

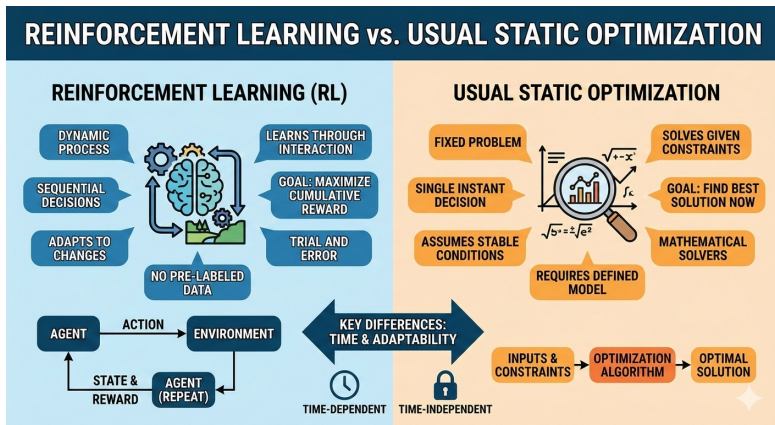
- **Static:** A "One-Time Diet." Assumes a perfect world with fixed inputs.
- **Dynamic (RL):** A "Smart Coach." Learns from mistakes and adapts to cravings.

OPTIMIZATION IN REAL LIFE: STATIC vs. DYNAMIC (RL), A HUMOROUS TAKE

HIGH RESOLUTION, BRIGHT COMIC-BOOK ART STYLE



RL vs. Usual Static Optimization



- **Static:** Mathematical solvers for fixed constraints.
- **RL:** Agent-Environment loop maximizing cumulative rewards.

Mastering Strategy: Reinforcement Learning in Chess

THE ROLE OF REINFORCEMENT LEARNING IN CHESS: MASTERING STRATEGY THROUGH SELF-PLAY

ROLE IN CHESS (THE “WHY”)



SUPERHUMAN PERFORMANCE

Achieves levels of play exceeding top human grandmasters by discovering novel strategies.



STRATEGIC INNOVATION

Discovers new openings, sacrifices, and long-term plans previously unknown to human theory.

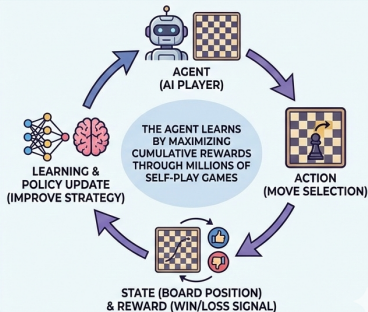


ULTIMATE TRAINING PARTNER

Provides human players with a powerful, tireless opponent to analyze and learn from.

Example: AlphaZero

HOW IT WORKS (THE RL CYCLE)



The Strategy

By using self-play, agents like AlphaZero discover novel sacrifices and long-term plans unknown to human theory.

Alpha Elimination: Reinforcing Numerical Algorithms

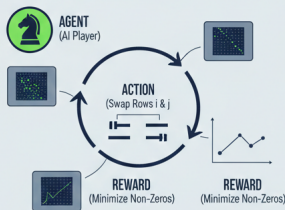
ALPHA ELIMINATION: REINFORCING THE ART OF ALGORITHMS

THE PROBLEM: MATRIX FILL-IN

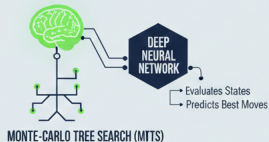
LU Decomposition creates "fill-in", making matrices dense and computations slow.



THE RL GAME: LEARNING OPTIMAL ORDERING



THE ALGORITHM'S BRAIN: MCTS & NEURAL NETWORKS



INPUT MASKING:

0s become 1, non-0,
non-0s become 1

THE IMPACT: SMARTER, FASTER LINEAR ALGEBRA



Achieves Superhuman Performance
Reduces Computational Cost & Memory.
Unlocks New Scientific Discovery

The Mathematical Objective

In Dynamic Optimization, we aim to find a policy π that maximizes the expected return G_t :

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Where:

- R is the reward signal.
- $\gamma \in [0, 1]$ is the discount factor.
- This differs from static $\min f(x)$ by accounting for the **future impact** of current decisions.

- **Efficiency:** RL reduces computational costs and memory in linear algebra.
- **Adaptability:** Handles time-dependent processes where static models fail.
- **Future:** Moving toward "Self-Optimizing" numerical libraries.

Thank you! Questions?